METRIC

MIL-PRF-62625C(AT)
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SUPERSEDING
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PERFORMANCE SPECIFICATION

VALVE, FUEL-DRAIN

This specification is approved for use by the U.S. Army Ground Vehicle Systems Center (GVSC), Combat Capabilities Development Command (CCDC), Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a check valve used for draining fuel.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to U.S. Ground Vehicle System Center, ATTN: FCDD-GVS-SAT, MS #267, 6501 E. 11 Mile Road, Detroit Arsenal, MI 48397-5000 or emailed to usarmy.detroit.ccdc-gvsc.mbx.standardization@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-PRF-7024 - Calibrating Fluids, Aircraft Fuel System Components

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

US ARMY TACOM DRAWINGS

12302407 - Valve, Fuel-Drain

(Copies of this drawing are available online at usarmy.detroit.ccdc-gvsc.mbx.standardization@mail.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL (ASTM)

ASTM D975 - Standard Specification for Diesel Fuel

(Copies of this document are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 First article. When specified (see 6.2), a first article sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specifications for valves. The materials shall be capable of meeting all of the operational and environmental requirements specified herein (see 4.4.1).

3.2.1 Metallic materials. Metallic materials shall be either corrosion resistant (such as passivated stainless steels, or anodized aluminum, etc.) or protected to prevent corrosion (using treatments such as organic coatings, metallizing treatments) (see 4.4.1).

3.2.2 Compatibility. The valve shall be capable of functioning with, and shall not deteriorate from contact with, the following fluids (see 4.4.1).

a. Turbine fuel per MIL-DTL-5624
b. Calibration fluid per MIL-PRF-7024
c. Diesel fuel oil per ASTM D975

3.2.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs (see 4.4.1).

3.3 Design and construction.

3.3.1 Envelope. The envelope of the valve shall be as specified on drawing 12302407.

3.4 Performance.

3.4.1 Open-close pressure. When pressure is applied in the closed flow path, the valve shall close between 55 kiloPascal (kPa) and 75 kPa (between 8 and 11 pounds per square inch gage (psig)). When the air pressure is decreased, the valve shall open between 75 and 55 kPa (between 11 and 8 psig) (see 4.4.4).

3.4.2 Proof pressure. The valve shall be capable of operating at 2068 kPa (300 psig) and shall be capable of withstanding a proof pressure of 3103 kPa (450 psig) with no signs of damage or external leakage (see 4.4.5).

3.4.3 Burst pressure. The valve shall be capable of withstanding a burst pressure of 5171 kPa (750 psig) without rupturing. The valve is not required to be operational after meeting this requirement (see 4.4.6).

3.4.4 Leakage. Leakage of calibration fluid conforming to MIL-PRF-7024, type II, or an equivalent, at 4137 ±138 kPa (600 ±20 psig) through the closed flow path shall be not more than 0.5 cubic centimeters per minute (cm^3/min). There shall be no external leakage (see 4.4.7).

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3.4.5 Drainage. The valve shall drain calibration fluid conforming to MIL-PRF-7024, type II, or an equivalent, at a rate of not less than 3500 cm^3/min with a pressure of 3.45 kPa (0.5 psig) (see 4.4.8).

3.4.6 Environmental. The valve shall not be damaged and shall meet the requirements of 3.4.1 to 3.4.5 during exposure to the following environments (see 4.4.9):

a. Ambient temperature range of -51 to 371 degrees Celsius (°C) (-60 to 700 degrees Fahrenheit (°F)).
b. Fluid temperature range of -51 to 371 °C (-60 to 700 °F).

3.5 Identification and markings. The valve shall be permanently marked for identification in accordance with Drawing 12302407 (see 4.4.3).

3.6 Workmanship. Valves shall be processed in such a manner as to be uniform in quality and free of defects that will affect their life, serviceability, or appearance (see 4.4.3).

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspections (see 4.3).
    1. Acceptance tests (see 4.3.2).
    2. Control tests (see 4.3.3).

4.2 First article inspection. First article inspection shall be performed on a minimum of three (3) valve samples produced under the production contract when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.3 (see table I) and the tests of table II.

4.3 Conformance inspection.

4.3.1 Sampling. Samples from an inspection lot for conformance inspections shall be selected in accordance with ANSI/ASQ Z1.4. Each sample selected shall be inspected as specified herein for the defects listed in table I. Conformance to a lot shall be accepted when zero (0) defects are found and rejected when one (1) or more defects are found. First article test samples shall not be shipped.

TABLE I. Classification of defects.

Category	Defect	Method of examination

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Critical	None	
Major:		
101	Valve, incomplete (see 3.3).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.3.1).	SIE 1/
103	Identification marking, improper (see 3.5).	Visual
104	Faulty workmanship affecting performance (see 3.6).	Visual
Minor:		
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3.1).	SIE 1/
202	Faulty workmanship affecting appearance (see 3.6).	Visual

1/ SIE = Standard Inspection Equipment.

TABLE II. Classification of inspections.

Title	Requirement	Inspection	First article	Conformance	
				Acceptance (100%)	Control
Materials and construction	3.2 thru 3.2.2	4.4.1	X		
Defects (see 4.2 and table I)	3.3.1, 3.5, and 3.6	4.4.3	X		
Open-close pressure	3.4.1	4.4.4	X	X	
Proof pressure	3.4.2	4.4.5		X	
Burst pressure	3.4.3	4.4.6	X		
Leakage	3.4.4	4.4.7	X		X
Drainage	3.4.5	4.4.8	X		X
Environmental	3.4.6	4.4.9	X		X

4.3.2 Acceptance tests (100 percent). Each valve selected in accordance with 4.3.1 shall be subjected to the tests specified in table II. Failure of any test shall be cause for rejection.

4.3.3 Control tests. Control tests shall be conducted on one valve from each lot of 200 units consecutively produced, except that not more than one test shall be performed in a three month period, nor less than one test in a six month period. The valves shall be subjected to the tests specified in table II.

4.4 Methods of inspection.

4.4.1 Materials. Conformance to 3.2 thru 3.2.2 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

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4.4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 21 ±10 °C (70 ±20 °F)
b. Relative humidity: 50 ±30 percent

4.4.3 Defects. Conformance to 3.3.1, 3.5 and 3.6 shall be determined by examination for the defects listed in table I. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.4.4 Open-close pressure test. To determine conformance to 3.4.1, submit the valve to increasing pressure and record when the valve poppet begins to close and when it fully closes. Then decrease the pressure, and record when the valve poppet begins to re-open and when it fully opens.

4.4.5 Proof pressure test. To determine conformance to 3.4.2, submit the valve to a pressure of 3103 kPa (450 psig) for 5 minutes, then examine it for damage or external leakage.

4.4.6 Burst pressure test. To determine conformance to 3.4.3, submit the valve to a pressure of 5171 kPa (750 psig), then examine it for signs of rupture.

4.4.7 Leakage test. To determine conformance to 3.4.4, subject the valve to a fluid pressure of 4137 ±138 kPa (600 ±20 psig), measure any leakage and record. Then examine the valve for external leakage.

4.4.8 Drainage test. To determine conformance to 3.4.5, submit the valve to a fluid pressure of 3.45 kPa (0.5 psig), and measure and record the rate of flow.

4.4.9 Environmental tests. To determine conformance to 3.4.6, submit the valve to the tests specified in 4.4.4 through 4.4.8 at the following environments:

a. Ambient and fluid temperatures of 371 °C (700 °F).
b. Ambient and fluid temperatures of -51 °C (-60 °F).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The valve covered by this specification is intended to be used for drainage from the combustor housing assembly.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. If required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2, and 2.3).
c. Title, number, and date of applicable drawings (see 2.2.2).
d. When first article is required (see 3.1).
e. Inspection conditions if other than as specified (see 4.4.2).
f. Packaging requirements (see 5.1).

6.3 Subject term (key word) listings.

Drainage
Draining
Fueling

6.4 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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